Universidade Estadual de Campinas
Instituto de Matemática, Estadística e Computaçao Científica
Programa de Pós-Graduaçao em Matemática

Notes of Group's Theory

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EL AMOR DE SU VIDA ⋆ Nestor Heli Aponte Avila[1]
Grupo Frontera & Grupo Firme n267452@dac.unicamp.br

** Content based on the course MM446 (Grupos e Representações) taught by Professor Plamen Kochloukov in the semester 2026-I, mainly following [1]. **

Notation ◇ Definition $(-)^{\diamond}$ Open $(-)^{\blacklozenge}$ Closed
□ Lemma ☐ Proposition ■ Theorem ⊠ Corollary

§ 1 Groups and Homomorphisms

Permutations

◇ Let $S_X := \{\alpha : X \to X \mid \alpha \text{ is and bijection}\}$ the set of all permutations of X and $* : S_X \times S_X \to S_X$ such that $\alpha * \beta := \alpha \circ \beta$. Usual notation:

$$X := \{1, 2, \ldots, n\} \text{ and } \alpha = \begin{pmatrix} 1 & 2 & \cdots & n \\ \alpha(1) & \alpha(2) & \cdots & \alpha(n) \end{pmatrix}.$$

◇ We call $\alpha \in S_n$ by *cycle* of length r if α fixes $n - r$ positions and $\alpha : i_1 \mapsto i_2 \mapsto \cdots \mapsto i_r \mapsto i_1$, in this case we write, $\alpha = \begin{pmatrix} i_1 & i_2 & \ldots & i_r \end{pmatrix}$. Every 1-cycle is the identity, $\mathbb{1}_X$, and 2-cycles are called by *transpositions*.

Example $S_3 \ni \alpha\beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1\ 3)(1\ 3\ 2) = (1)(2\ 3).$

◇ Two cycles $\alpha, \beta \in S_X$ are *disjoint* when they affect different elements. In other words, $\{x \in X : \alpha(x) \neq x \text{ and } \beta(x) \neq x\} = \varnothing$.

■ **1.1.** Every $\alpha \in S_n$ is either a cycle or a product of disjoint cycles.

The proof is by induction on k, points moved by α. Given α, let i_1 moved by α, construct a cycle $\sigma = (i_1 \;\; i_2 \;\; \dots \;\; i_r)$, conclude that $\alpha = \sigma\alpha'$, with σ and α' disjoint, how α' moved fewer positions than α, the result follows.

◇ A *complete factorization* of α is an expresion of α as a product of disjoint cycles that includes 1-cycles for every j fixed by α.

■ **1.2.** Let $\alpha \in S_n$. A complete factorization $\alpha = \beta_1\beta_2 \dots \beta_k$ is unique except for the order in which the factors occur.

Suppose $\alpha = \gamma_1\gamma_2 \dots \gamma_s$ a second complete factorization. If β_t moves i_1, then $\exists\gamma_j$ that moves i_1, let $j = s$ (Ex. 8). Now, $\forall k \in \mathbb{N}$, $\beta_t^k(i_1) = \alpha^k(i_1) = \gamma_s^k(i_1)$ (Ex. 14), so $\beta_t = \gamma_s$ and $\beta_1 \dots \beta_{s-1} = \gamma_1 \dots \gamma_{t-1}$ (Ex. 6), the result follows by induction on $\max\{s, t\}$.

■ **1.3.** Every $\alpha \in S_n$ is a product of transpositions.

By ■ 1.1. it suffices to factor cycles: $(1 \;\; 2 \;\; \dots \;\; r) = (1 \;\; r)(1 \;\; r-1) \cdots (1 \;\; 2)$.

◇ For a permutation $S_n \ni \alpha = \beta_1 \dots \beta_t$ (complete factorization), the *signum* of α is $\operatorname{sgn}(\alpha) := (-1)^{n-t}$. *Even* or *odd* depends on the quantities of transpositions in the decomposition of α (in traspositions).

■ **1.7.** For all $\alpha, \beta \in S_n$, $\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta)$. Also,

i. α is *even* $\leftrightharpoons \operatorname{sgn}(\alpha) = 1$.

ii. α is *odd* $\leftrightharpoons$ it's product of odd transpositions.

→ Doesn't support summaries, look if you want [1, pag. 9]

Groups

◇ A *group* $(G, *)$ is a nonemptyset G, equipped with an associative binary operation $* : G \times G \to G$ such that:

G1. $\exists e \in G : \forall a \in G,\ a * e = e * a = a$. → Identity element

G2. $\forall a \in G,\ \exists b \in G :\ a * b = b * a = e$. → Existence of inverses

If, in addition, the operation $*$ is commutative, then G is called an *abelian group*.

Examples • $(S_n, \circ)$ the *symetric group*. | • $(\mathbb{F}, +)$, usual addition in fields.
• $(\mathbb{Z}_n, +)$, where $\mathbb{Z}_n := \{[a] \in \mathbb{Z}/\sim : a \sim b \leftrightharpoons a \equiv b \pmod{n}\}$ • $(\mathrm{GL}_n(\mathbb{F}), \cdot)$

■ **1.10.** If G is a group, the identity and the inverse of each element are unique. $\rightarrow$ It's simply, suppose another verifying the propertie as always

⊠ **1.11.** We denote the inverse of a by a^{-1}. It follows inmediately that $(a^{-1})^{-1} = a$.

◇ Given $a \in G,\ n \in \mathbb{N}$, we define $a^n := \overbrace{a * a * \cdots * a}^{\text{n times}}$. We also set $a^0 := e$ and $a^{-n} := (a^{-1})^n$, all this expresions are *powers* of a.

Homomorphisms

◇ Let $(G, *)$ and $(H, \circ)$ be groups. A function $f : G \to H$ is a *homomorphism* if $\forall a, b \in G$, $f(a * b) = f(a) \circ f(b)$; If f is a bijection, $f : G \cong H$ (*isomorphism*).

■ **1.13.** Let $f : (G, *) \to (G', \circ)$ homomorphism. Then,

i. $f(e) = e'$. ii. $f(a^{-1}) = f(a)^{-1}$. iii. $f(a^n) = f(a)^n$.

For (i) and (ii) apply f to the equations $e = e * e$ and $a * a^{-1} = e = a^{-1} * a$. The affirmation (iii) follows by induction in n and (ii).

§ 2 Isomorphism Theorems

Subgroups

◇ A nonemptyset $S \subset G$ is a *subgroup* of G if (i) $s \in S$ implies $s^{-1} \in S$ and (ii) $s, t \in G$ imply $st \in G$. We denote $S \leq G$.

■ **2.1.** $(S, *) \leq (G, *)$ is a group in its own right. $\rightarrow *|_{S \times S}$ + definition

■ **2.2.** $S \leq G, \leftrightharpoons \mathbb{1} \in S$ and $s, t \in S$ imply $st^{-1} \in S$.

◇ Given $a \in G,\ \{a^m : m \in \mathbb{Z}\} =: \langle a \rangle \leq G$ is the *cyclic subgroup generated by* a; G is *cyclic* if $\exists a : G = \langle a \rangle$ and $|\langle a \rangle|$ is the *order* of a.

■ **2.3.** If $|\langle a \rangle| = m < \infty$, then $m = \min\{k : a^k = \mathbb{1},\ k \in \mathbb{N}\}$.

It's clear that $\exists k \in \mathbb{N} : a^k = \mathbb{1}$. Let $\{a^j\}_{j=0}^{k-1} : \forall i < l \leq k-1,\ a^i \neq a^l$, prove that $a^k = \mathbb{1}$ (by contradiction). Finally prove $\langle a \rangle = \{a^j\}$ using the division algorithm.

⊠ **2.4.** If $|G| < \infty$, then $S \leq G \ \leftrightharpoons\ s, t \in S$ imply $st \in S$. ($\Leftarrow$) If $s \in S$, then $\langle s \rangle \leq S$, $\exists m \in \mathbb{N} : s^m = \mathbb{1}$ and $s^{-1} = s^{m-1} \in S$

Note. Clearly $\{\mathbb{1}\} < H < G \leq G$, in this case H is a *proper subgroup*.

Kernel and Image of $f : G \to H$ homomorphism

- $\{a \in G : f(a) = \mathbb{1}_H\} =: \ker f \leq G$;
- $\{h \in H : h = f(a),\ a \in G\} =: \operatorname{Im} f \leq H$.

■ **2.5.** Let $(S_i)_{i \in I} : S_i \leq G$, then $\bigcap S_i \leq G$. $\to$ Mogollas

◇ Given $X \subset G$, $\langle X \rangle := \bigcap \{S : S \leq G \text{ and } X \subset S\}$ is the subgroup of G generated by X. If $|X| = n < \infty$, we write $\langle X \rangle = \langle a_1, a_2, \ldots, a_n \rangle$.

Note. If $H, K \leq G$, then we denote $\langle H \cup K \rangle$ as $H \vee K$.

◇ If $\varnothing \neq X \subset G$, then a *word* on X is an expression of the form $x_1^{e_1} x_2^{e_2} \ldots x_n^{e_n} = w \in G$, where $x_j \in X$ and $e_j = \pm 1$.

■ **2.7.** $\langle X \rangle \leq G$ is precisely the set of all words on X. $\to$ Let W the set of all words, prove that $W \leq G$ is the smallest : $X \subset W$

Lagrange's Theorem

◇ Let $S \leq G$ and $t \in G$. The *right coset* of S in G is the subset $St := \{st : s \in S\}$, we call t a *representative* of St (similarly we define tS).

Example $\langle n \rangle =: S < \mathbb{Z}$. Given $a \in \mathbb{Z},\ a + S = \{a + qn : q \in \mathbb{Z}\} = [a] \in \mathbb{Z}_n$.

Note. In general tS and St need not coincide, may be different.

□ **2.8.** $Sa = Sb$ (or $aS = bS$) $\leftrightharpoons\ ab^{-1} \in S$ (or $b^{-1}a \in S$).

($\Rightarrow$) $\mathbb{1}a \in Sa = Sb$, so $\exists s \in S : a = sb \ \leftrightharpoons\ ab^{-1} = s \in S$. ($\Leftarrow$) Prove two inclusions using $Sa \ni x = sa = s\sigma b \in Sb : \sigma = ab^{-1}$ (similarly for the other).

■ **2.9.** For any two right (or two left) cosets of S in G are either identical or disjoint.
$\to$ If $x = sa = tb$, then $ab^{-1} = s^{-1}t \in S$, result follows from ■ 2.8.

■ **2.10.** The number of right cosets is equal to the number of left cosets of S in G.

Proof that $f : \mathscr{L} \to \mathscr{R}$ such that $Sa \mapsto a^{-1}S$ is a well defined bijection.

◇ The *index* of S in G, denoted by $[G : S]$, is the number of right cosets.

■ **2.11.** (***Lagrange***) If $|G| < \infty$ and $S \leq G$, then $|S|$ divides $|G|$ and $[G : S] = {}^{|G|}/_{|S|}$.

By ■ 2.9. $G = St_1 \sqcup St_2 \sqcup \ldots \sqcup St_n$, and so $|G| = \sum |St_j|$, how $f_j : S \mapsto St_j$ is bijection, $|St_j| = |S|$ and $|G| = n|S|$, where $n = [G : S]$.

⊠ **2.12.** If $|G| < \infty$ and $a \in G$, then the order of a divides $|G|$. Moreover, if $|G| = p$ prime, then $|G|$ is a cyclic group. $\to$ Takes $a \neq \mathbb{1}$, then $|\langle a \rangle|$ divides $p \leftrightharpoons |\langle a \rangle| = p$

⊠ **2.14.** (***Fermat***) If p is prime and $a \in \mathbb{Z}$, then $a^p \equiv a \pmod p$.

Let $\mathbb{F} = \mathbb{Z}_p$ and $G = (\mathbb{F}^\times, \cdot)$. If $[a] = [0]$, then clearly $[a]^p = [a] = [0]$. Otherwise, $[a]^{p-1} = [\mathbb{1}]$, so $[a]^p = [a]$, any case $a^p \equiv a \pmod p$.

Cyclic Groups

□ **2.15.** If $G = \langle a \rangle : |G| = n < \infty$, then $\exists! S \leq G : |S| = d$ divides n.

Let $H = \langle b \rangle : |H| = d$ (Ex. 18), then $b^d = \mathbb{1} = (a^m)^d$, so $\exists k \in \mathbb{N} : md = nk$ (Ex. 13). Finally, $b = a^m = (a^{n/d})^k$ and $H \leq \langle a^{n/d} \rangle =: S \leq G$ (Ex. 11). Since $|S| = d$, we have $H = S$.

◇ $\varphi(n) := \#\{k \in \mathbb{N} : k < n \text{ and } (k, n) = 1\}$.

■ **2.16.** If n is a positive integer, then $n = \sum_{d \mid n} \varphi(d)$.

Let $G = \langle a \rangle : |G| = n$, then $G = \bigsqcup \operatorname{gen} C : C \leq G$. By □ 2.15. $\forall d \in \operatorname{Div}(n)$, $\exists! C_d \leq G$, so $|G| = \sum_{d \mid n} |\operatorname{gen} C_d| = \sum_{d \mid n} \varphi(d)$ (Ex. 20).

■ **2.17.** $G : |G| = n$ is cyclic $\leftrightharpoons \forall d : d \mid n,\ \exists! C_d \leq G : |C_d| = d$.

($\Rightarrow$) □ 2.15.. ($\Leftarrow$) By ■ 2.16. $n = \sum |\operatorname{gen} C| \leq \sum \varphi(d) = n$, so G must have a subgroup of order $n \leftrightharpoons G$ is cyclic.

■ **2.18.** $G \leq \mathbb{F}^\times : |G| < \infty$ is cyclic. In particular, $\mathbb{F}^\times$ is cyclic if $|\mathbb{F}| < \infty$. $\rightarrow$ The polynomial $x^d - \mathbb{1}$ has almost d roots in $\mathbb{F}[x]$, result follows from ■ 2.17.

■ **2.19.** Let p prime. A group $G : |G| = p^n$ is cyclic $\leftrightharpoons$ it's an abelian group and $\exists! S \leq G : |S| = p$. $\rightarrow$ Look [1, pag. 29], needs (Ex. 13)

Normal Subgroups

$$\diamond \quad ST := \{st : s \in S,\ t \in T\} \subset G.$$

■ **2.20.** (***Product Formula***) If $S, T \leq G : |G| < \infty$, then $|ST||S \cap T| = |S||T|$. [1] $\rightarrow$ Doesn't support sumaries, look if you want [1, pag. 30]

$\diamond$ $K \leq G$ is a *normal subgroup*, denoted by $K \lhd G$, if $\forall g \in G,\ gKg^{-1} = K$.

Example If $f : G \to H$ is homomorphism, then $\ker f \lhd G$.

Note. $K \lhd G \leftrightharpoons Kg = gK$, so there's a *partial commutativity*, if $g \in G$ and $k \in K$, $\exists k' \in K : gk = k'g$.

$\diamond$ A *conjugate* of $x \in G$ is an elemente of the form $axa^{-1} : a \in G$.

■ **2.21.** If $N \lhd G$, then the cosets form a group, the *quotient group*, denoted by G/N, of order $[G : N]$. $\rightarrow$ Set $Na * Nb := NaNb$ and prove the properties

⊠ **2.22.** If $N \lhd G$, then the *natural map*, $\pi : G \to G/N$ such that $a \mapsto Na$, is a surjective homomorphism with $\ker \pi = N$. $\rightarrow$ It's clear

Note. The quotient is the generalization of the construction of $\mathbb{Z}_n$ as $\mathbb{Z}/\langle n \rangle$ and all normal subgroup is kernel of some homomorphism.

$\diamond$ Given $a, b \in G$, we denote the *commutator* as $[a, b] := aba^{-1}b^{-1}$ and the *commutator subgroup* as G', the generated by all the commutators.

Example $X := \{[a, b] : a, b \in G\}$ need not to be a subgroup as $[a, b] * [c, d]$ need not to be a commutator (Ex. 43).

■ **2.23.** $G' \lhd G$. Moreover, if $H \lhd G$, then G/H is abelian $\leftrightharpoons G' \leq H$.

[1] ST need not to be a subgroup.

If $f : G \to G$ is homomorphism, then $f([a,b]) = [f(a), f(b)]$ so $G' \lhd G$ (Ex. 34); Moreover, $Hab = Hba \leftrightharpoons (ab)(ba)^{-1} = aba^{-1}b^{-1} = [a,b] \in H$.

The Isomorphism Theorems

■ **2.24.** (***First Isomorphism Theorem***) Let $f : G \to H$ homomorphism with $\ker f = K$. Then $K \lhd G$ and $\exists \varphi : {}^{G}/_{K} \cong \operatorname{Im} f$. $\to$ Let $\varphi : Ka \mapsto f(a)$ and prove

□ **2.25.** If $S, T \leq G$ and $S \lhd G$ (or $T \lhd G$), then $ST = S \vee T = TS$.

($\subseteq$) It's clear; ($\supseteq$) Follows if $ST, TS \leq G$, because of note after ■ 2.5.; If $T \lhd G$, $s_j \in S$, $t_j \in T$, show that $s_1 t_1 (s_2 t_2)^{-1} \in ST$.

■ **2.26.** (***Second Isomorphism Theorem***) Let $T, N \leq G$ with N normal. Then $N \cap T \lhd T$ and ${}^{T}/_{N \cap T} \cong {}^{NT}/_{N}$. $\to$ By ■ 2.24. (*FIT*) with $\ker \pi|_T = N \cap T \lhd T$ and ${}^{T}/_{N \cap T} \cong \operatorname{Im} \pi|_T = {}^{NT}/_{N}$

■ **2.27.** (***Third Isomorphism Theorem***) Let $K \leq H$ where both $K, H \lhd G$. Then ${}^{H}/_{K} \lhd {}^{G}/_{K}$ and $({}^{G}/_{K}) \,/\, ({}^{H}/_{K}) \cong {}^{G}/_{H}$. $\to$ Follows by ■ 2.24. (*FIT*). with $f : {}^{G}/_{K} \twoheadrightarrow {}^{G}/_{H} \mid Ka \mapsto Ha$ and $\ker f = {}^{H}/_{K}$

■ **2.28.** (***Correspondence Theorem***) Let $K \lhd G$ and $\pi : G \to {}^{G}/_{K}$. Then $S \mapsto \pi(S) = {}^{S}/_{K}$ is a bijection between $\{S : K \leq S \leq G\}$ and $\{H : H \leq {}^{G}/_{K}\}$. Moreover, if we denonte $S^* = {}^{S}/_{K}$, then:

i. $T \leq S \leftrightharpoons T^* \leq S^*$, and then $[S : T] = [S^* : T^*]$.

ii. $T \lhd S \leftrightharpoons T^* \lhd S^*$, and then ${}^{S}/_{T} = {}^{S^*}/_{T^*}$.

$\to$ Doesn't support summaries, see [1, pag. 38]

◇ A group $\mathbb{1} < G$ is *simple* if $\nexists K \lhd G : \mathbb{1} < K < G$.

Direct Products

◇ Given H and K groups, their *direct product*, denoted by $H \times K$ is a group with the operation (by coordinates), $* : (h,k) \times (h',k') \mapsto (hh', kk')$.

Note. $H \cong H \times \mathbb{1}_K$ and $K \cong \mathbb{1}_H \times K$.

■ **2.29.** Let $H, K \lhd G$. If $HK = G$ and $H \cap K = \mathbb{1}$, then $G \cong H \times K$.

Let $f : G \to H \times K$ such that $a = hk \mapsto (h, k)$, use commutators to see that $h'kh'^{-1}k^{-1} \in H \cap K$ and prove the remaining properties.

All hypothesis in ■ 2.29. are necessary

Example $G = S_3$ (non abelian), $H = \langle(1\ \ 2\ \ 3)\rangle$ and $K = \langle(1\ \ 2)\rangle$. Check everything except by $K \ntrianglelefteq G$. We have $H \times K \cong \mathbb{Z}_3 \times \mathbb{Z}_2$ (abelian).

■ **2.30.** If $A \lhd H$ and $B \lhd K$, then $A \times B \lhd H \times K$ and ${}^{H\times K}/_{A\times B} \cong {}^{H}/_{A} \times {}^{K}/_{B}$.

Follows by ■ 2.24. *(FIT)* with $\varphi : H \times K \twoheadrightarrow {}^{H}/_{A} \times {}^{K}/_{B}$ such that $(h, k) \mapsto (Ah, Bk)$ and $\ker \varphi = A \times B$.

⊠ **2.31.** If $G = H \times K$, then ${}^{G}/_{H\times\mathbb{1}} \cong {}^{H}/_{H} \times {}^{K}/_{\mathbb{1}} \cong K$.

§ 3 Symmetric Groups and G-Sets

Conjugates

□ **3.1.** The relation (in G) $x \sim y \leftrightharpoons \exists g \in G : y = gxg^{-1}$ is an equivalence relation.

◇ ⋆ $a^G := [a] \in {}^{G}/_{\sim}$ ← *Conjugacy classes*

⋆ $\{a \in G : a^G = a\} =: Z(G) \lhd G$ ← *Center* (is abelian)

⋆ $\{g \in G : ag = ga\} =: C_G(a) \leq G$ ← *Centralizer* of $a \in G$

■ **3.2.** $|a^G| = [G : C_G(a)]$, divisor in case that $|G| < \infty$.

Let $C := C_G(a)$ and consider $f : a^G \to {}^{G}/_{C}$ such that $gag^{-1} \mapsto gC$. Prove that f is a well defined bijection. Second affirmation follows by ■ 2.11. *(Lagrange)*.

◇ Given $H \leq G$ and $g \in G$ we have similar definitions:

⋆ $gHg^{-1} = \{ghg^{-1} : h \in H\} =: H^g \leq G$ ← *Conjugate*

⋆ $\{a \in G : aHa^{-1} = H\} =: N_G(H) \leq G$ ← *Normalizer* of H in G

Note. Notice that $H \lhd N_G(H) \leq G$ is the largest which verifies this property.

■ **3.3.** $\#\{H^g : g \in G\} = [G : N_G(H)]$, divisor in case $|G| < \infty$. Moreover, $H^a = H^b \leftrightharpoons b^{-1}a \in N_G(H)$.

> Let $[H] := \{H^g : g \in G\}$ and $N := N_G(H)$. Prove that $f : [H] \to {}^G/_N$ such that $H^a \mapsto aN$ is a well defined bijection.

Symmetric Groups

◇ Two permutations $\alpha, \beta \in S_n$ have the same *cycle structure* if their complete factorizations have the same number of r-cycles for each r.

□ **3.4.** If α and β have the same cycle structure, then $\pi := \alpha\beta\alpha^{-1}$ is obtain by applying α to the symbols in β.

> If β fixes i, then $\pi(\alpha(i)) = \alpha(i)$. Let $\beta = \gamma_1\gamma_2\cdots(\ldots i \;\; j \ldots)\cdots\gamma_t$, $k = \alpha(i)$, $l = \alpha(j)$ and concludes that $\pi : k \mapsto l$.

■ **3.5.** $\alpha \sim \beta \leftrightharpoons$ they have the same cycle structure.

> ($\Rightarrow$) If $\alpha = \gamma_1\gamma_2\cdots(\ldots i \;\; j \ldots)\cdots\gamma_t$ and $\beta = \delta_1\delta_2\cdots(\ldots k \;\; l \ldots)\cdots\delta_t$, define η such that $i \mapsto k$ and $j \mapsto l$. Result follows by □ 3.4..

⊠ **3.6.** $H \lhd S_n \leftrightharpoons \alpha \in H$ implies $\beta \in H$, $\forall \beta : \alpha \sim \beta$. → (Ex. 2.34)

◇ $\{\alpha \in S_n : \alpha \text{ is even}\} =: A_n \leq S_n$.

■ **3.7.** $A_4 \leq S_4$ is a group of order 12 having no subgroup of order 6.

> If $\exists S \leq A_4 : |S| = 6$, then $[G : S] = 2$ and $\alpha^2 \in S$, $\forall \alpha \in A_4$ (Ex. 2.16). We have $\alpha = \alpha^4 = (\alpha^2)^2$, so every 3-cycle (eight in total) lies to S.↯

■ **3.11.** A_n is simple for $n \geq 5$. → Doesn't support sumaries, see [1, pag. 50]

Some Representation Theorems

■ **3.12.** (***Cayley, 1878***) Every group G can be imbedded as a subgroup of S_G. In particular, if $|G| = n < \infty$, then $\exists L : G \cong \operatorname{Im} L \leq S_n$ (*regular representation*).

> Left translations $L_a : G \to G$ such that $x \mapsto ax$ are bijections (Ex. 1.33), so $L_a \in S_G$. Prove that $L : a \mapsto L_a$ is an injective homomorphism.

⊠ **3.13.** If $G : |G| = n < \infty$ and $\mathbb{F}$ is a field, then $\exists \iota : G \cong H \leq \mathrm{GL}_n(\mathbb{F})$. $\to$ Use $\mathrm{PL}_n(\mathbb{F}) := \{A \in \mathrm{GL}_n(\mathbb{F}) : a_{ij} \in \{0,1\}\} \cong S_n$

■ **3.14.** If $H \leq G$ and $[G : H] = n$, then $\exists \rho : G \to S_n$ homomorphism such that $\ker \rho \leq H$ (*representation in cosets*).

> Let $X := {}^{G}/_{H}$ and $\rho_a : X \to X$ such that $\rho_a : gH \mapsto agH$. Verify that $\rho_a \in S_X$ and $\rho : a \mapsto \rho_a$ is an homomorphism that satisfies this kernel property.

⊠ **3.15.** If G is simple and $\exists H \leq G : [G : H] = n > 1$, then $\exists \rho : G \cong \operatorname{Im} \rho \leq S_n$. $\to$ $\exists \rho : G \to S_n$ with $\ker \rho \lhd H \leq G \;\leftrightharpoons\; \ker \rho = \mathbb{1}$ (ρ is injective)

⊠ **3.16.** An infinite simple group G has no proper subgroups of finite index.

■ **3.17.** Let $H \leq G$ and $X := \{H^g : g \in G\}$. There is a homomorphism $\psi : G \to S_X$ with $\ker \psi \leq N_G(H)$ (*representation in conjugates*).

> Let $\psi_a : X \to X$ such that $\psi_a : H^g \mapsto (H^g)^a$. Verify that $\phi_a \in S_X$ and so $\psi : a \mapsto \psi_a$ is homomorphism with this kernel property.

G-Sets

◇ If X is a set and G is a group, then X is a *G-set* if $\exists \alpha : X \times G \to X$ (called an *action*), denoted by $\alpha : (g, x) \mapsto gx$ such that:

(A1) $\mathbb{1}x = x,\ \forall x \in X.$ (A2) $g(hx) = (gh)x,\ \forall g, h \in G,\ \forall x \in X.$

Example $G \leq S_X$ and $\alpha = \mathrm{ev} : G \times X \to X$ such that $\mathrm{ev} : (\sigma, x) \mapsto \sigma(x) = \sigma x$.

■ **3.18.** If X is a G-set with action α, then $\exists \widetilde{\alpha} : G \to S_X$ homomorphism given by $\widetilde{\alpha}(g) : x \mapsto gx = \alpha(g, x)$. Conversely, every homomorphism $\varphi : G \to S_X$ defines an action, namely, $gx = \varphi(g)x$ which makes X into G-set.

◇ Given X a G-set and $x \in X$, we define:

- ⋆ $\mathcal{O}(x) := \{gx : g \in G\} \subset X \quad \leftarrow$ *G-orbit* of x
- ⋆ $G_x := \{g \in G : gx = x\} \leq G \quad \leftarrow$ *Stabilizer* of x

G acts on himself by conjugation

- $\mathcal{O}(x) = [x] \in {}^{G}/_{\sim}$ and $G_x = C_G(x)$.
- $\mathcal{O}(H) = \{H^g : g \in G\}$ and $G_H = N_G(H)$.

■ **3.19.** If X is a G-set, then $|\mathcal{O}(x)| = [G : G_x]$. $\to$ Let $f : \mathcal{O}(x) \to {}^{G}/_{G_x} \mid ax \mapsto aG_x$ and prove that is a well defined bijection

Dihedral (or Symmetric) Group

Example Let $\mathbb{1} = X := \{z \in \mathbb{C} : z^{2n} = 1\}$, $n \geq 2$ and consider: s (*counterclockwise rotation* of X) and t (*reflection* respect real axis). Notice that $D_{2n} := \langle s, t \rangle$ is a group such that:

$$\text{(i)}\ \ s^n = \mathbb{1}; \qquad \text{(ii)}\ \ t^2 = \mathbb{1}; \qquad \text{(iii)}\ \ tst = s^{-1}.$$

■ **3.32.** If $|G| < \infty$ and $a, b \in G$ are such that $|a| = |b| = 2$, then $\langle a, b \rangle \cong D_{2n}$ for some $n \in \mathbb{N}$. $\to$ See [1, pag. 68]

§ 4 The Sylow Theorems (G finito)

p-Groups

◇ If p is prime, then G is a *p-group* if $\exists a \in G : |a| = p$.

□ **4.1.** If G is abelian such that $|G| = pm$, then G is a p-group.

By induction on m, let $x \in G : |x| = t > 1$. If $p \mid t$, $|x^{t/p}| = p$ (Ex. 2.11), other case $\langle x \rangle \lhd G$ and $G^* := {}^{G}/_{\langle x \rangle}$ is abelian : $|G^*| = pm'$, $m' < m$ so $\exists a^* = \pi(a) \in G^* : |a^*| = p$, we have $|a| = pk$ (Ex. 2.14) and $|a^k| = p$.

■ **4.2.** If $|G| = pm$, then $\exists a : |a| = p$.

If $\exists x \in G \setminus Z(G) : p$ divides $|C_G(x)|$, result follows by induction. Other case p divides $[G : C_G(x)]$ and choosing suitable representatives we have:

$$\circledast \qquad |G| = |Z(G)| + \sum [G : C_G(x_j)] \quad \leftarrow \textit{Class equation} \text{ of } G$$

So p divides $|Z(G)|$ (abelian) and result follows by □ 4.1..

⊠ **4.3.** G is p-group $\leftrightharpoons$ $|G| = p^k$. $\to$ Mogollas

■ **4.4.** If $\mathbb{1} < G$ is a p-group, then $\mathbb{1} < Z(G)$. $\to$ By ⊠ 4.3. every $C_G(x_i)$ is a p-group, so from ⊛ $|Z(G)| = p^k,\ k > 1$

⊠ **4.5.** If $|G| = p^2$, then G is abelian. $\to$ If $|Z(G)| = p$ then $G/Z(G) \cong \mathbb{Z}_p$ cyclic.↯ (Ex. 3.3), so $|Z(G)| = p^2$ and $G = Z(G)$ abelian

■ **4.6.** If $|G| = p^k$, then (i) If $H < G$, then $H < N_G(H)$ and (ii) Every $H \leq G$ maximal is normal and $[G : H] = p$. $\to$ See [1, pag. 75] and (Ex. 2.58)

■ **4.8.** If $|G| = p^k$ and $r_s := \#\{H \leq G : |H| = p^s\}$, then $r_s \equiv 1 \pmod p$. $\to$ Doesn't support summaries, see [1, pag. 76]

The Sylow Theorems

◇ $P \leq G$ is a *p-Sylow subgroup* if is a maximal p-subgroup of G.

□ **4.11.** If $P \leq G$ is p-Sylow, then (i) $(|N_G(P)/P|, p) = 1$ and (ii) If $|a| = p^k$ and $P^a = P$, then $a \in P$. $\to$ See [1, pag. 78]

■ **4.12.** ***(Sylow)*** Let $P \leq G$ p-Sylow and $n_p := \#\{H \leq G : H \text{ is } p\text{-Sylow}\}$, then

i. $\forall H \leq G$ p-Sylow, $\exists g \in G : H = P^g$.

ii. $n_p \equiv 1 \pmod p$ and $n_p \mid |G|$.

$\to$ Doesn't support summaries, see [1, pag. 79]

⊠ **4.13.** $\exists! P \leq G$ p-Sylow $\leftrightharpoons$ $P \lhd G$. $\to$ Mogollas

■ **4.14.** If $|G| = p^e m$, where $(p, m) = 1$, then $\forall P \leq G$ p-Sylow, $|P| = p^e$.

> $[G : P] = [G : N_G(P)][N_G(P) : P]$, where $[G : N_G(P)] = n_p \equiv 1 \pmod p$ while $[N_G(P) : P] = |N_G(P)/P|$ prime to p by □ 4.11.. By ■ 2.11. (*Lagrange*) $|P| = p^k,\ k \leq e$ and $[G : P] = |G|/|P| = p^{e-k}m$, but is prime to p so $k = e$.

⊠ **4.15.** If p^k divides $|G|$, then $\exists H \leq G : |H| = p^k$. $\to$ Let $P \leq G$ p-Sylow, then p^k divides $|P|$ and result follows (Ex. 2)

References

[1] Rotman, J. J. (1984). *An introduction to the theory of groups*. Boston: Allyn and Bacon.